

Optional challenge exercises

- Write an uptime program that prints the uptime in terms of ticks using the uptime system call.

```
FatimaFaisal@xv6:~$ cd xv6-labs-2025
FatimaFaisal@xv6:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_
ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forp
han user/_dorphan user/_uptime user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1023 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -
drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ uptime
uptime
uptime: 27 ticks
$ uptime
uptime
uptime: 48 ticks
$ uptime
uptime
uptime: 78 ticks
$ uptime
uptime
uptime: 102 ticks
$ uptime
uptime
uptime: 163 ticks
$ |
```

- Support regular expressions in name matching for find. grep.c has some primitive support for regular expressions.

```
FatimaFaisal@xv6:~$ cd xv6-labs-2025
FatimaFaisal@xv6:~/xv6-labs-2025$ git branch
* firstAssignment
  problemSet1
  util
FatimaFaisal@xv6:~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user
/_kill user/_ln user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie
user/_logstress user/_forphan user/_dorphan user/_uptime user/_find user/_memdump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1023 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-l
egacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . sh
find . sh
./findtest.sh
./sh
$ find . .*\.sh
find . .*\.sh
$ find . .*\.c
find . .*\.c
$ find . ^sh.*
```

```

find . ^sh.*
./sh
$ find . .*
find . .*
./README
./findtest.sh
./sixfive.txt
./cat
./echo
./forktest
./grep
./init
./kill
./ln
./ls
./mkdir
./rm
./sh
./stressfs
./usertests
./grind
./wc
./zombie
./logstress
./forphan
./dorphan
./uptime
./find
./memdump
$

```

- The xv6 shell (user/sh.c) is just another user program. It lacks many features found in real shells, but you can modify and improve it. For example, modify the shell to not print a \$ when processing shell commands from a file ([moderate](#)), modify the shell to support wait ([easy](#)), modify the shell to support tab completion ([easy](#)), modify the shell to keep a history of passed shell commands ([moderate](#)), or anything else you would like your shell to do. (If you are very ambitious, you may have to modify the kernel to support the kernel features you need; xv6 doesn't support much.)

```

rm          2 14 34056
sh          2 15 58984
stressfs    2 16 34944
usertests   2 17 184264
grind       2 18 50264
wc          2 19 36120
zombie      2 20 33424
logstress   2 21 35968
forphan     2 22 34848
dorphan     2 23 34288
uptime      2 24 33704
find        2 25 43472
sleep       2 26 33888
sixfive     2 27 35760
memdump     2 28 40832
memdump     2 28 40832
console     3 30 0
$ echo hello
hello
$ echo "history"
"history"
$ echo "hello"
"hello"
$ history
0: ls
1: echo hello
2: echo "history"
3: echo "hello"

```